New Lower Bounds for Erdős–Szekeres Numbers in Three and Four Dimensions

Deep Bhattacharjee [1, 2, *] and Ushashi Bhattacharya [3, †]

[1] *Formerly, Electro-Gravitational Space Propulsion Laboratory (EGSPL), Bhubaneswar, Odisha 751030, India*
[2] *Formerly, University of Calcutta, Senate House, 87/1 College Street, Kolkata 700073, West Bengal, India*
[3] *Formerly, National Taiwan University, Taipei 106319, Taiwan*

(Dated: October 19, 2026)

We construct eleven points in general position in four-dimensional space with no eight of them in convex position, and fifteen points in general position in three-dimensional space with no eight of them in convex position. The first set raises the lower bound for the least number of points in four-space that forces eight in convex position from eleven to twelve, so that this number is now known to be twelve or thirteen. The second raises the corresponding lower bound in three-space from fourteen to sixteen. Both sets have small integer coordinates, and each proof is a finite list of identities that place one point of the set strictly inside a simplex spanned by others.

2020 Mathematics Subject Classification. 52C10; 52A37; 05D10.

Keywords: Erdős–Szekeres problem; convex position; points in higher dimensions; interior-point certificates

CONTENTS

I. Introduction 1
II. Relations and certificates 3
III. Eleven points in four-space 4
IV. Fifteen points in three-space 5
V. How the sets were found 5
VI. What remains open 6
Data and code availability 6
Ethical declarations 6
Author contributions 6
A. Proofs of the lemmas 6
B. The certificates 7
References 7

I. INTRODUCTION

A finite set of points in $\mathbb{R}^d$ is in *general position* if no $d+1$ of its points lie on a common hyperplane, and in *convex position* if none of its points lies in the convex hull of the others. A set of k points in convex position is a *k-gon*; a k-gon of a set X is a *k-hole* of X if its convex hull contains no other point of X.

In 1935 Erdős and Szekeres [1] proved that for every k there is a least integer $g^{(2)}(k)$ such that every set of $g^{(2)}(k)$ points in general position in the plane contains a k-gon.[1] The value $g^{(2)}(5)=9$ goes back to Makai [2]. Szekeres and Peters proved $g^{(2)}(6)=17$ with a computer search [3]; Marić gave a formally verified proof [4], and Scheucher a proof by satisfiability solving [7]. In 2017 Suk proved $g^{(2)}(k)\le 2^{k+o(k)}$ [5], so the conjecture holds up to the error term in the exponent. For holes the answer changes: there are arbitrarily large sets in general position in the plane without 7-holes [2], while Heule and Scheucher proved that every 30 points in general position contain a 6-hole and that 29 points do not suffice [6].

The theorem holds in every dimension. Let $g^{(d)}(k)$ be the least integer such that every set of $g^{(d)}(k)$ points in general position in $\mathbb{R}^d$ contains a k-gon, and let $h^{(d)}(k)$ be the corresponding number for k-holes, possibly infinite. Projection onto a generic plane shows $g^{(d)}(k)\le g^{(2)}(k)$ [2], and Károlyi's dimension reduction [8] gives the sharper estimate $g^{(d)}(k)\le g^{(d-1)}(k-1)+1$. Károlyi and Valtr [9] constructed sets showing $g^{(d)}(k)\ge c_d^{\,k^{1/(d-1)}}$ for a constant $c_d>1$, and Pohoata and Zakharov [10] proved $g^{(d)}(k)=2^{o(k)}$ for every $d\ge 3$, so in three and more dimensions the growth is not even known up to the exponent. Valtr [11] showed that sets without large holes exist in every dimension.

The small values form a short list (Table I). Trivially $g^{(d)}(k)=k$ for $k\le d+1$, and $g^{(d)}(d+2)=d+3$ by a theorem of Motzkin recorded by Danzer, Grünbaum and Klee

* itsdeep@live.com; Corresponding author. Second address: d.bhattacharjee@erl-forschung.de.
† rai.bhattacharya.in@gmail.com

[1] The question came from Esther Klein, who had observed that five points in general position in the plane always contain the vertices of a convex quadrilateral; she married Szekeres soon after, and the result became known as the happy ending problem. The paper [1] contains two proofs, one through Ramsey's theorem and one through cups and caps, and the second gives $g^{(2)}(k)\le\binom{2k-4}{k-2}+1$. Erdős and Szekeres conjectured $g^{(2)}(k)=2^{k-2}+1$ and later constructed 2^{k-2} points without a k-gon, which shows that the conjectured value cannot be lowered. The survey of Morris and Soltan [2] tells this history and collects the improvements of the upper bound up to 2000.

TABLE I. Values and bounds of $g^{(d)}(k)$ for $3 \le d \le 5$ and $5 \le k \le 9$. Entries in bold are proved in this paper; the earlier lower bounds were 11, 14 and 15. The upper bounds 13 are Scheucher's [7].

	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$
$d=3$	6	9	13	$\ge$ **16**	$\ge$ **17**
$d=4$	5	7	9	**12**–13	≥ 13
$d=5$	5	6	8	10	12–13

[12]; see also [17]. For $d+2 \le k \le \lfloor 3d/2 \rfloor + 1$,

$$g^{(d)}(k) = 2k - d - 1, \tag{1}$$

where the lower bound is a construction of Bisztriczky and Harborth [13], the upper bound is due to Bisztriczky and Soltan [14], and the two are combined in [2, Theorem 4.1]. Beyond this range Bisztriczky and Soltan proved $g^{(3)}(6) = 9$ [14]. Morris and Soltan observed that all values known in 2000 with $k > \lfloor (3d+1)/2 \rfloor$ satisfy $g^{(d)}(k) = 4g^{(d)}(k-d) - 3$ and conjectured this for all such k [2, Conjecture 4.2]. In 2023 Scheucher [7] encoded the problem as a satisfiability question for acyclic chirotopes, the combinatorial abstraction of point sets, and proved

$$g^{(3)}(7) = 13, \qquad g^{(4)}(8) \le 13, \qquad g^{(5)}(9) \le 13, \tag{2}$$

together with the same upper bounds for holes. His twelve points in $\mathbb{R}^3$ without a 7-gon are integer points. His twelve-element witnesses in rank five and six are chirotopes that he proved to be non-realizable by points, and he conjectured that realizable witnesses exist. So the lower bounds $g^{(4)}(8) \ge 11$ and $g^{(5)}(9) \ge 12$ given by (1) remained in place, and in $\mathbb{R}^3$ the only lower bound for $g^{(3)}(8)$ was $g^{(3)}(7)+1 = 14$.[2]

We prove the following.

Theorem A. *There is a set of eleven points in general position in $\mathbb{R}^4$ with no eight points in convex position. Consequently $g^{(4)}(8) \ge 12$ and $h^{(4)}(8) \ge 12$, and by (2) both numbers lie in $\{12, 13\}$.*

Theorem B. *There is a set of fifteen points in general position in $\mathbb{R}^3$ with no eight points in convex position. Consequently $g^{(3)}(8) \ge 16$, $h^{(3)}(8) \ge 16$ and $g^{(3)}(9) \ge 17$.*

Theorem A is the first lower bound beyond (1) in four dimensions, and it leaves two candidates for $g^{(4)}(8)$; the conjecture of Morris and Soltan and Scheucher's conjecture both predict 13. Theorem B raises the lower bound for $g^{(3)}(8)$ by two; the conjecture of Morris and Soltan predicts $g^{(3)}(8) = 4g^{(3)}(5) - 3 = 21$, and the best upper bound we know comes from dimension reduction and the planar estimate of Tóth and Valtr, as in [7].

[2] If X has no k-gon and y is any point with $X \cup \{y\}$ in general position, then $X \cup \{y\}$ has no $(k+1)$-gon, because removing y from a $(k+1)$-gon leaves a k-gon of X. Hence $g^{(d)}(k+1) \ge g^{(d)}(k)+1$. The same argument applies to holes, since the convex hull of a subset of a hole contains no point of the set outside the hole.

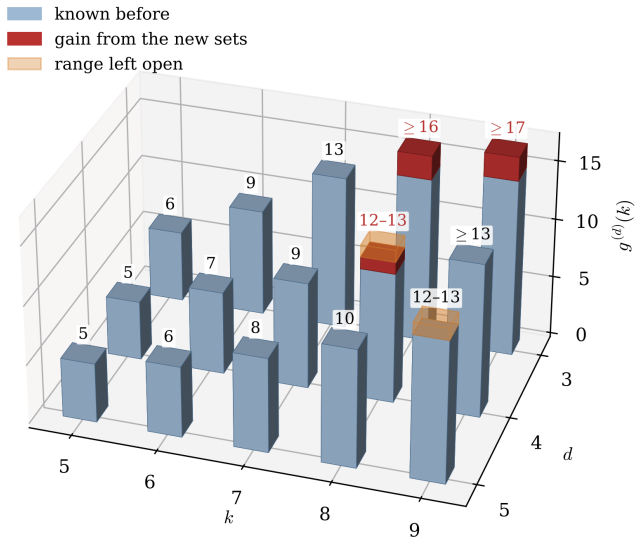


FIG. 1. The values of $g^{(d)}(k)$ for $d = 3,4,5$ and $k = 5,\dots,9$. Blue bars are the values or lower bounds known before; red bars show the gain from Theorems A and B; the light orange blocks mark the ranges left open below Scheucher's upper bounds. Where no upper bound is drawn, only the lower bound is shown.

Both proofs are certificates. If $v_0, \dots, v_d$ are affinely independent points of $\mathbb{R}^d$ and

$$\sum_{j=0}^{d} m_j v_j = Mp, \qquad m_j \in \{1,2,3,\dots\}, \quad M = \sum_{j=0}^{d} m_j, \tag{3}$$

then p lies in the interior of the simplex spanned by the v_j, and no set containing p and the v_j is in convex position. We call (3) a *relation*. A set has no k-gon as soon as every k of its points contain the $d+2$ points of some relation (Proposition II.5). Twenty-eight relations prove Theorem A and 161 prove Theorem B; they are listed in Appendix B. Each relation is an identity between integer vectors that can be checked by hand. That the relations meet every 8-subset is a finite combinatorial condition: for Theorem A it is a statement about 165 triples, which Table IV lets the reader check directly, and for Theorem B it concerns 6435 subsets and is checked by the programs of the archive. Since relations survive small perturbations (Lemma II.4), general position costs nothing; the integer sets we give are in general position as they stand.

The two sets have a layered structure (Figures 3 and 4). The eleven points in $\mathbb{R}^4$ consist of the six vertices of a bipyramid over a tetrahedron and five points inside it, and every relation places one of the five inner points in a simplex. The fifteen points in $\mathbb{R}^3$ form three convex layers, a triangular bipyramid, an octahedron and a tetrahedron, and no relation places a vertex of the outer bipyramid inside a simplex. Scheucher's twelve points in $\mathbb{R}^3$ without a 7-gon have convex layers of sizes 5, 6 and 1, again a triangular bipyramid and an octahedron. For other extremal problems on explicit configurations in dimensions four and seven see [18, 19], and for reflexive polytopes in dimension four see [20].

Section II proves the certificate criterion, Sections III and IV prove Theorems A and B, Section V explains how the sets were found, and Section VI discusses what remains open. Appendix A contains the detailed proofs of the lemmas of Section II, and Appendix B the two certificates.

II. RELATIONS AND CERTIFICATES

Throughout, X is a finite set in $\mathbb{R}^d$ and conv, int denote the convex hull and the interior. For points $v_0,\dots,v_d$ write $\det(v_0,\dots,v_d)$ for the determinant of the $(d+1)\times(d+1)$ matrix whose rows are $(1,v_j)$; it vanishes exactly when the points lie on a common hyperplane, and its absolute value is $d!$ times the volume of the simplex they span.

Lemma II.1. *Let X be in general position and $Y\subseteq X$. Then Y is in convex position if and only if no point of Y lies in* $\operatorname{int}\operatorname{conv}\sigma$ *for a set σ of $d+1$ other points of Y.*

Proof. If $p\in\operatorname{int}\operatorname{conv}\sigma$ with $\sigma\subseteq Y\setminus\{p\}$, then $p\in\operatorname{conv}(Y\setminus\{p\})$ and Y is not in convex position. Conversely, suppose $p\in\operatorname{conv}(Y\setminus\{p\})$. By Carathéodory's theorem [17] p is a convex combination of affinely independent points $v_0,\dots,v_r$ of $Y\setminus\{p\}$, and we choose r minimal, so all coefficients are positive. If $r<d$, then p and $v_0,\dots,v_r$ are $r+2\le d+1$ points on an r-dimensional affine subspace, hence on a hyperplane, contrary to general position. So $r=d$, the simplex $\operatorname{conv}\{v_0,\dots,v_d\}$ is full-dimensional, and a point with positive barycentric coordinates lies in its interior. □

Definition II.2. A *relation* of X is a triple (p,σ,m) where $p\in X$, $\sigma=\{v_0,\dots,v_d\}\subseteq X\setminus\{p\}$ satisfies $\det(v_0,\dots,v_d)\neq 0$, and $m=(m_0,\dots,m_d)$ consists of positive integers with $\sum_j m_j v_j = Mp$, $M=\sum_j m_j$. Its *support* is $\{p\}\cup\sigma$, a set of $d+2$ points.

Lemma II.3. *If (p,σ,m) is a relation, then $p\in\operatorname{int}\operatorname{conv}\sigma$, the barycentric coordinates of p with respect to σ are m_j/M, and if the points have integer coordinates and* $\gcd(m_0,\dots,m_d)=1$, *then M divides* $\det(v_0,\dots,v_d)$.

Proof. The v_j are affinely independent, so every point has unique barycentric coordinates with respect to them, and (3) shows that those of p are $m_j/M>0$. Write $D=\det(v_0,\dots,v_d)$ and let N_j be the same determinant with v_j replaced by p. By Cramer's rule the jth barycentric coordinate of p is N_j/D (Appendix A), so $m_jD=MN_j$ for every j. The right-hand sides are integer multiples of M, and an integer combination of the m_j equals 1, hence M divides $\det(v_0,\dots,v_d)$. □

In the tables of the appendices we record, for each relation, the integer $g=|\det(v_0,\dots,v_d)|/M$. A reader who checks one determinant per relation thereby also checks $\det\neq 0$; Lemma II.3 explains why g is always an integer.

Lemma II.4. *Let (p,σ,m) be a relation of X. There is $\varepsilon>0$ such that for every map $x\mapsto x'$ with $|x'-x|<\varepsilon$ on the support, p' lies in the interior of the simplex spanned by the images of σ.*

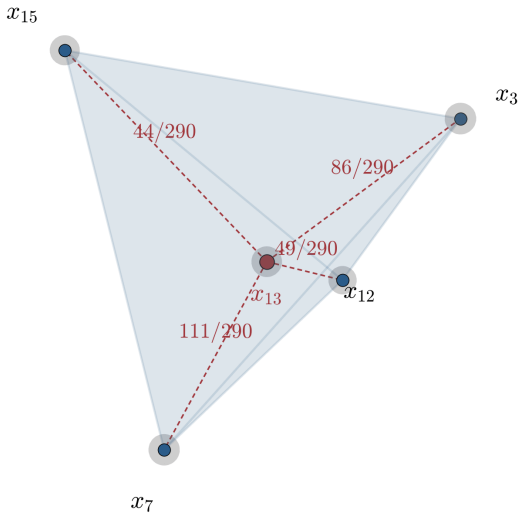


FIG. 2. One relation of the set of Theorem B, drawn after an affine map that turns the simplex into a regular tetrahedron and keeps barycentric coordinates. The point p is the weighted mean of the four vertices with the positive integer weights shown. The small balls indicate that every point may move a little without p leaving the simplex (Lemma II.4).

Proof. The barycentric coordinates of p with respect to σ are rational functions of the coordinates of the $d+2$ points whose common denominator is $\det(v_0,\dots,v_d)$. They are therefore continuous where this determinant does not vanish, and at the given points they are positive. Continuity gives an ε that keeps the determinant nonzero and the coordinates positive. Appendix A gives an explicit ε. □

Proposition II.5. *Let X be a finite set in $\mathbb{R}^d$ and $\mathscr{R}$ a set of relations of X such that every k-subset of X contains the support of some relation in $\mathscr{R}$. Then X contains no k points in convex position. Moreover, there is a set X' in general position, obtained from X by moving each point by an arbitrarily small amount, with the same property.*

Proof. Let $Y\subseteq X$ have k points and let (p,σ,m) be a relation with support in Y. By Lemma II.3, $p\in\operatorname{conv}\sigma\subseteq\operatorname{conv}(Y\setminus\{p\})$, so Y is not in convex position. For the second statement let ε be the least of the constants of Lemma II.4 over the finitely many relations of $\mathscr{R}$. The configurations of $|X|$ points in $\mathbb{R}^d$ that are not in general position form the zero set of the product of the polynomials $\det(x_{i_0},\dots,x_{i_d})$, none of which vanishes identically, so they form a closed set with empty interior in $\mathbb{R}^{d|X|}$. Hence there are points x' with $|x'-x|<\varepsilon$ in general position. Every relation of $\mathscr{R}$ still places p' in the interior of the image simplex, so the first argument applies to X'. □

The condition of Proposition II.5 can be stated through complements. A k-subset Y contains the support S of a relation if

TABLE II. The eleven points of Theorem A in $\mathbb{R}^4$.

point	x	y	z	w
x_1	0	19	34	30
x_2	2	36	25	0
x_3	18	0	19	1
x_4	19	18	40	32
x_5	20	32	0	23
x_6	23	31	15	1
x_7	12	16	24	14
x_8	13	27	21	13
x_9	16	24	14	17
x_{10}	16	25	19	25
x_{11}	18	24	20	14

and only if $X \setminus Y \subseteq X \setminus S$. So if $|X| = n$, the sets $X \setminus S$ of size $n-d-2$ must cover every $(n-k)$-subset of X. This is a covering design, and Schönheim's bound [15] gives a lower bound for the number of relations any such certificate needs.

Remark II.6. A certificate in the sense of Proposition II.5 for eleven points in $\mathbb{R}^4$ and $k = 8$ covers the 3-subsets of an 11-set by 5-sets, and needs at least $\lceil \frac{11}{5} \lceil \frac{10}{4} \lceil \frac{9}{3} \rceil \rceil \rceil = 18$ relations. For fifteen points in $\mathbb{R}^3$ and $k = 8$ it covers the 7-subsets of a 15-set by 10-sets, and needs at least 74 relations. Our certificates use 28 and 161 relations.

By Lemma II.1 the method is complete: if a set in general position has no k-gon, then every k-subset contains a point in the interior of a simplex of other points of the subset, and when the coordinates are rational the barycentric coordinates are rational, so the relation can be written with integer weights. A certificate exists exactly when the set has no k-gon.

III. ELEVEN POINTS IN FOUR-SPACE

Let $X_4 = \{x_1, \dots, x_{11}\} \subset \mathbb{R}^4$ be the points of Table II. All coordinates lie between 0 and 40.

Proposition III.1. *The convex hull of X_4 is a bipyramid over a tetrahedron: its vertices are $x_1, \dots, x_6$, and its eight facets are the tetrahedra spanned by x_1 or x_6 together with three of x_2, x_3, x_4, x_5. The points $x_7, \dots, x_{11}$ lie in the interior.*

Proof. For each of the eight listed quadruples τ the determinants $\det(\tau, x)$ have one sign for all seven remaining points x of X_4, which is a sign check of 56 determinants. So the hyperplane through τ supports $\operatorname{conv} X_4$, and since τ consists of four affinely independent points it spans a facet. Each triangle contained in one of these eight tetrahedra lies in exactly two of them. In a convex polytope every ridge lies in exactly two facets and the facets are connected through ridges [16], so these eight tetrahedra are all the facets of $\operatorname{conv} X_4$, and they are the facets of the bipyramid $\operatorname{conv}\{x_1, \dots, x_6\}$. The points $x_7, \dots, x_{11}$ lie strictly on the inner side of all eight facet hyperplanes, by the same sign check, so they are interior. □

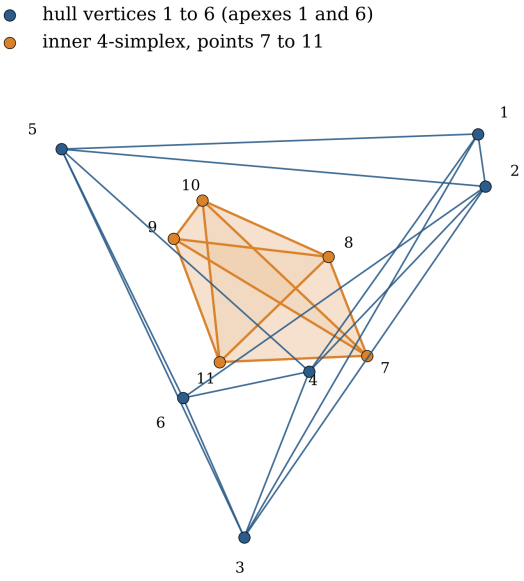


FIG. 3. A schematic view of the set of Theorem A, projected to three dimensions after an affine map that makes the hull a regular bipyramid. The tetrahedron $x_2x_3x_4x_5$ and the apexes x_1, x_6 span the hull; the five inner points (red) form a 4-simplex, drawn enlarged. Every one of the 28 relations of Table IV places a red point inside a simplex of other points.

Proof of Theorem A. Each row of Table IV is a relation of X_4: the identity $\sum_j m_j x_{s_j} = M x_p$ holds in each of the four coordinates, the weights are positive, and the determinant of the simplex is $\pm gM \neq 0$. The last column of the table lists, for each relation, the five points outside its support. Every one of the $\binom{11}{3} = 165$ triples of indices lies in one of these 28 five-element sets. So the complement of every 8-subset of X_4 lies outside the support of some relation, that is, every 8-subset contains a support, and Proposition II.5 shows that X_4 has no 8 points in convex position. The set X_4 is itself in general position: the $\binom{11}{5} = 462$ determinants of five of its points are all nonzero, the smallest in absolute value being 11. Independently of this check, Proposition II.5 provides a perturbation of X_4 in general position with the same property. Hence $g^{(4)}(8) \geq 12$. An 8-hole is an 8-gon, so $h^{(4)}(8) \geq g^{(4)}(8) \geq 12$, and Scheucher's bound $h^{(4)}(8) \leq 13$ [7] together with $g^{(4)}(8) \leq h^{(4)}(8)$ gives the last statement. □

Every relation places one of the inner points $x_7, \dots, x_{11}$ in a simplex; the outer points are vertices of the hull and lie inside nothing. Any eight points of X_4 include at least two inner points, and the certificate shows that one of the inner points among them always lies inside a simplex spanned by four or five of the others.

TABLE III. The fifteen points of Theorem B in $\mathbb{R}^3$.

point	x	y	z
x_1	0	128	0
x_2	53	0	130
x_3	92	166	46
x_4	110	180	108
x_5	171	139	79
x_6	73	88	95
x_7	75	161	71
x_8	79	143	55
x_9	98	169	67
x_{10}	110	176	104
x_{11}	123	128	82
x_{12}	81	109	93
x_{13}	85	154	67
x_{14}	86	154	66
x_{15}	101	163	69

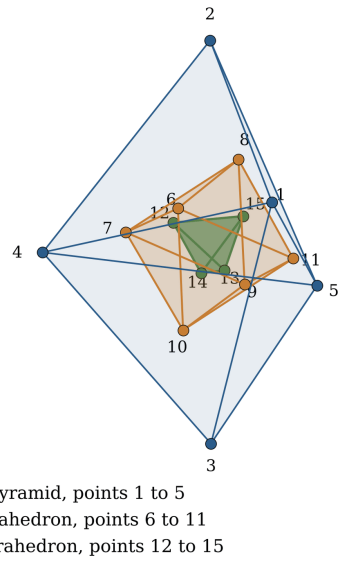


FIG. 4. The three convex layers of the set of Theorem B: a triangular bipyramid (blue), an octahedron (green) and a tetrahedron (red), with the labels of Table III. The drawing is schematic: it shows regular solids with the true combinatorics and labels, since in the actual coordinates the inner layers are small and flat.

IV. FIFTEEN POINTS IN THREE-SPACE

Let $X_3 = \{x_1, \dots, x_{15}\} \subset \mathbb{R}^3$ be the points of Table III. All coordinates lie between 0 and 180.

Proposition IV.1. *The set X_3 has three convex layers. The vertices of* $\operatorname{conv} X_3$ *are $x_1, \dots, x_5$, forming a triangular bipyramid with apexes x_2 and x_3 over the triangle $x_1x_4x_5$. The vertices of the convex hull of the remaining points are $x_6, \dots, x_{11}$, forming an octahedron whose opposite vertices are x_6, x_9, x_7, x_{11} and x_8, x_{10}. The points $x_{12}, \dots, x_{15}$ span a tetrahedron inside it.*

Proof. As for Proposition III.1. The six triangles of the bipyramid have constant sign of $\det(\tau, x)$ over the other twelve points of X_3, so they are facets of $\operatorname{conv} X_3$; every edge among them lies in exactly two, so they are all the facets, and the remaining ten points lie strictly inside. The eight triangles of the octahedron have constant sign over the other seven points of $\{x_6, \dots, x_{15}\}$ and close up in the same way, so they are the facets of the second layer, with $x_{12}, \dots, x_{15}$ strictly inside. Four points in general position span a tetrahedron. □

Proof of Theorem B. Each row of Tables V and VI is a relation of X_3, by the same three checks as before. Every one of the $\binom{15}{8} = 6435$ eight-point subsets of X_3 contains the support of one of these 161 relations; equivalently, every 7-subset of the indices lies in the complement of a support. This finite statement is verified in the archive [22] by independent programs in four languages and by the Lean 4 kernel; three of the programs also test every 8-subset directly for convex position in exact integer arithmetic, without using the certificate. Proposition II.5 then shows that X_3 has no 8 points in convex position. The set X_3 is in general position, the smallest absolute value of its $\binom{15}{4} = 1365$ determinants being 6, and again Proposition II.5 would supply a perturbation if this check were omitted. So $g^{(3)}(8) \geq 16$ and $h^{(3)}(8) \geq 16$, and the monotonicity $g^{(3)}(9) \geq g^{(3)}(8) + 1$ gives $g^{(3)}(9) \geq 17$. □

The points p of the relations run through all ten points of the two inner layers, and no relation places a vertex of the outer bipyramid inside a simplex. In 90 of the 161 relations the point p is a vertex of the inner tetrahedron. The others are needed because an 8-subset may avoid the inner tetrahedron altogether: it then consists of eight of the eleven points of the two outer layers, and one of its octahedron vertices must lie inside a simplex of the others.

V. HOW THE SETS WERE FOUND

The sets were found by local search over coordinates. For a candidate set we counted its k-subsets in convex position, but a bare count gives the search nothing to follow, since most moves change it by zero. We therefore scored each k-subset Y by how far it is from failing: for every point p of Y and every simplex σ of other points of Y, the least barycentric coordinate of p with respect to σ measures how far p is from being inside, and the score of Y is f of the smallest such violation over p and σ, with $f(t) = t/(1+t)$ for violations and 0 for subsets that already fail. The total score is zero exactly for sets without k-gons. Simulated annealing on this score, with occasional *repair moves* that take a point of a surviving k-gon and place it just inside the simplex of the subset nearest to it,

found the eleven points in $\mathbb{R}^4$ and the fifteen points in $\mathbb{R}^3$.[3]

The sets so found have real coordinates and some relations with very small barycentric coordinates. We then *polished* each set by hill climbing on the least barycentric coordinate over a fixed certificate, which doubled it for the set in $\mathbb{R}^3$, and rounded to the coarsest integer grid on which a certificate survives; this gave coordinates at most 40 in $\mathbb{R}^4$ and at most 180 in $\mathbb{R}^3$. Finally we chose a certificate with few relations and small weights by a randomised greedy covering followed by removal of redundant relations. None of these steps is part of the proof.

VI. WHAT REMAINS OPEN

The value of $g^{(4)}(8)$ is now 12 or 13. The same search, run on twelve points in $\mathbb{R}^4$, ended with sets that still contain four 8-gons, and twelve points in $\mathbb{R}^5$ without a 9-gon ended at three 9-gons; for fifteen points in $\mathbb{R}^3$ the search reached zero, while sixteen points ended at four 8-gons. Scheucher's twelve-element chirotopes without 8-gons and 9-gons in rank five and six are not realizable either. A twelve-point set in $\mathbb{R}^4$ without an 8-gon would prove $g^{(4)}(8) = 13$ and confirm both conjectures quoted in the introduction; a proof that every twelve points contain an 8-gon would refute them.

In three dimensions the gap is wider: $16 \le g^{(3)}(8)$, while the conjectured value is 21. The layered structure of X_3 suggests adding a fourth layer inside the tetrahedron or replacing the tetrahedron by a larger polytope. The value of $h^{(3)}(8)$ is not even known to be finite [7].

Finally, the certificates are longer than the bounds of Remark II.6: 28 relations against 18 in $\mathbb{R}^4$, and 161 against 74 in $\mathbb{R}^3$. In both sets the points p of the relations are confined to the inner layers, so the supports are not spread evenly over the set, as a covering design meeting Schönheim's bound would require.

DATA AND CODE AVAILABILITY

The coordinates, the certificates, the search, polishing and rounding programs, the independent checks in Python, C, C++ and Julia, the Lean 4 file that checks both certificates in the kernel, the figure programs and the LaTeX source are in the repository `github.com/creelie/Erdos-s1` and its Zenodo archive [22]. The status check of the problem was made with the BLLM engine Anaxia, release v3.0.5 [21]. Deep Bhattacharjee carried out the computations and the Python, C, C++, Julia, Lean and shell verifications with AI assistance (Claude, Anthropic).

ETHICAL DECLARATIONS

Funding: none. Competing interests: none.

AUTHOR CONTRIBUTIONS

Deep Bhattacharjee proposed the problem, carried out the searches and computations and wrote the paper. Ushashi Bhattacharya checked the relations of Table IV and the proofs of Appendix A. Both authors approved the final version.

Appendix A: Proofs of the lemmas

Barycentric coordinates. Let $v_0, \dots, v_d \in \mathbb{R}^d$ with $D = \det(v_0, \dots, v_d) \neq 0$. For a point p the system $\sum_j \lambda_j = 1$, $\sum_j \lambda_j v_j = p$ has the matrix whose columns are $(1, v_j)$, the transpose of the matrix defining D, so it has the unique solution

$$\lambda_j = \frac{\det(v_0, \dots, v_{j-1}, p, v_{j+1}, \dots, v_d)}{D} \tag{A1}$$

by Cramer's rule. The point p lies in the interior of $\operatorname{conv}\{v_0, \dots, v_d\}$ if and only if all $\lambda_j > 0$: the simplex is the set where all $\lambda_j \ge 0$, and its boundary is where some $\lambda_j = 0$. Summing (A1) gives $\sum_j \det(v_0, \dots, p, \dots, v_d) = D$, an identity used to check tables: the numerators of a relation are $m_j D/M$ and add up to D.

Proof of the divisibility in Lemma II.3. From $\lambda_j = m_j/M$ and (A1), $m_j D = M N_j$ with N_j the integer numerator. Choose integers a_j with $\sum_j a_j m_j = 1$. Then $D = \sum_j a_j m_j D = M \sum_j a_j N_j$.

An explicit ε in Lemma II.4. Let all $d+2$ points lie in a ball of radius $R \ge 1$ around the origin and let $\delta = \min_j m_j/M$. The determinant $\det(v_0, \dots, v_d)$ is a multilinear function of the rows $(1, v_j)$. Moving each point by at most $\varepsilon \le R$ changes each row by at most ε and keeps every row of norm at most $\sqrt{1+4R^2} \le 3R$. Replacing the rows one at a time and applying Hadamard's inequality to each difference, the determinant changes by at most $(d+1)\,\varepsilon\,(3R)^d$, and the same holds for each numerator in (A1). If $\varepsilon \le |D|\,\delta / \big(4(d+1)(3R)^d\big)$, then the new determinant has the sign of D and modulus at least $\frac{3}{4}|D|$, each new numerator differs from the old one $\lambda_j D$ by at most $\frac{1}{4}\delta|D|$, and so every new barycentric coordinate is at least $\frac{3}{4}\delta/\frac{5}{4} = \frac{3}{5}\delta > 0$.

The general position claim of Proposition II.5. Each $\det(x_{i_0}, \dots, x_{i_d})$ is a polynomial in the $d|X|$ coordinates that

[3] Before these constructions we checked the status of the problem in the literature cited here and with the BLLM engine Anaxia [21]. The engine keeps indexes of DeepMind's formal-conjectures collection and AlphaEvolve problem repository, of the OpenRefute archive and of OpenAI's collection of mathematical manuscripts, and answers a query with the matching entries and their recorded status. For the Erdős–Szekeres problem these indexes hold only the planar question (Erdős problem 107 in the formal-conjectures collection and problem 58 of the AlphaEvolve repository) and no entry on $g^{(d)}(k)$ for $d \ge 3$, which agreed with our reading of [7] and of the papers citing it. The two point sets of this paper were found and verified by the search and certificate programs of the archive [22], not by the engine. The methods that found them, a graded count of violated subsets, repair moves that push a point just inside its nearest simplex, and the polishing of the least barycentric coordinate described below, were afterwards added to the engine as construction rules.

TABLE IV. The twenty-eight relations behind Theorem A. In each row the point x_p satisfies $\sum_j m_j x_{s_j} = M x_p$ with $\sigma = \{s_1, \dots, s_5\}$ listed in increasing order, the weights m_j are positive integers with sum M, and $|\det \sigma| = gM$ for the 5×5 determinant of the rows $(1, x_s)$, $s \in \sigma$. The last column lists the five points not used by the relation; every triple of $\{1, \dots, 11\}$ lies in one of these sets.

p	σ	m_1	m_2	m_3	m_4	m_5	M	g	unused points
7	1,2,3,4,10	7673	6278	15495	2902	10393	42741	9	5,6,8,9,11
7	1,2,3,6,11	1878	304	1910	74	2217	6383	3	4,5,8,9,10
7	1,3,4,8,9	30287	42693	6085	50075	3654	132794	1	2,5,6,10,11
7	1,3,6,8,11	614	550	194	152	297	1807	6	2,4,5,9,10
8	1,2,3,6,10	1410	2694	598	3109	4628	12439	9	4,5,7,9,11
8	1,2,5,6,7	44866	11396	12870	62375	5226	136733	1	3,4,9,10,11
8	1,2,5,6,9	9820	2570	1833	13526	2613	30362	2	3,4,7,10,11
8	1,2,5,10,11	2589	9511	2031	3027	22996	40154	1	3,4,6,7,9
8	1,2,9,10,11	723	6103	2031	3933	11728	24518	1	3,4,5,6,7
8	2,3,4,5,11	6507	1387	3142	3765	3887	18688	9	1,6,7,9,10
8	2,3,6,10,11	3114	328	109	3458	4230	11239	3	1,4,5,7,9
8	2,4,5,9,11	11222	3936	1011	12483	11060	39712	1	1,3,6,7,10
8	2,4,6,7,10	1990	104	1315	1442	3186	8037	18	1,3,5,9,11
8	2,7,9,10,11	1380	241	228	1214	2219	5282	3	1,3,4,5,6
9	1,3,5,6,11	19212	13094	38497	14620	12438	97861	1	2,4,7,8,10
9	1,5,6,7,10	1198	19329	4916	23140	2066	50649	1	2,3,4,8,11
9	2,3,4,5,7	1574	1628	1448	6753	3144	14547	12	1,6,8,10,11
9	2,3,4,5,10	14378	20588	3032	37755	37728	113481	1	1,6,7,8,11
9	2,3,5,7,11	290	104	4493	4616	2172	11675	8	1,4,6,8,10
9	3,5,7,8,10	1726	6667	3702	5104	2392	19591	1	1,2,4,6,11
10	1,4,5,6,9	14386	23140	33623	3752	9577	84478	1	2,3,7,8,11
10	1,4,5,7,8	14270	30691	47880	6133	10247	109221	1	2,3,6,9,11
10	1,4,5,9,11	4022	5844	9829	11	3752	23458	1	2,3,6,7,8
11	1,3,4,6,10	3149	4165	2999	16899	15406	42618	6	2,5,7,8,9
11	1,4,6,7,9	16	898	1756	853	2136	5659	13	2,3,5,8,10
11	3,4,6,7,10	608	1150	5848	3149	5748	16503	6	1,2,5,8,9
11	3,4,6,8,9	5074	16942	25719	14465	36606	98806	1	1,2,5,7,10
11	4,5,6,7,10	280	228	686	639	166	1999	16	1,2,3,8,9

does not vanish identically, since it is nonzero on any set of $d+1$ affinely independent points. A finite union of zero sets of nonzero polynomials has empty interior, so every neighbourhood of a configuration contains configurations in general position.

Schönheim's bound. A family of s-subsets of an n-set covering every t-subset has at least $\lceil \frac{n}{s} \lceil \frac{n-1}{s-1} \cdots \lceil \frac{n-t+1}{s-t+1} \rceil \cdots \rceil \rceil$ members [15]: the members containing a fixed point, with that point removed, cover all $(t-1)$-subsets of the remaining $n-1$ points, and counting incidences gives the recursion. For $(n,s,t) = (11,5,3)$ the bound is $\lceil \frac{11}{5} \lceil \frac{10}{4} \cdot 3 \rceil \rceil = \lceil \frac{11 \cdot 8}{5} \rceil = 18$; for $(15,10,7)$ the nested values are $3, 6, 11, 19, 31, 49, 74$.

Appendix B: The certificates

Table IV lists the 28 relations of X_4, and Tables V and VI the 161 relations of X_3, ordered by p and then by σ. The weights are the integers $m_j = M\lambda_j$ with $\gcd(m_0, \dots, m_d) = 1$; the largest M is 136,733 for X_4 and 261,355 for X_3. To check a row, multiply each point x_{s_j} by m_j, add, and compare with Mx_p coordinate by coordinate; then compute one determinant of size $d+1$, whose absolute value must be gM.

[1] P. Erdős and G. Szekeres, A combinatorial problem in geometry, *Compositio Math.* **2** (1935), 463–470, http://www.numdam.org/item/CM_1935__2__463_0/.

[2] W. Morris and V. Soltan, The Erdős–Szekeres problem on points in convex position—a survey, *Bull. Amer. Math. Soc.* **37** (2000), 437–458, doi:10.1090/S0273-0979-00-00877-6.

TABLE V. The 161 relations behind Theorem B. Each row gives $\sum_j m_j x_{s_j} = M x_p$ with $\sigma = \{s_1, \ldots, s_4\}$ in increasing order, positive integer weights m_j with sum M, and $|\det \sigma| = gM$ for the 4×4 determinant of the rows $(1, x_s)$.

p	σ	m_1	m_2	m_3	m_4	M	g	p	σ	m_1	m_2	m_3	m_4	M	g
6	1,2,3,12	1939	18838	7809	46242	74828	3	9	3,4,10,14	88	1	49	6	144	24
6	1,2,10,11	9628	26527	16482	12471	65108	16	9	3,4,14,15	383	192	25	98	698	12
6	2,3,4,8	34648	2701	12313	31010	80672	1	9	3,5,10,13	17959	6	10191	1152	29308	3
6	2,3,7,9	37155	8333	10899	22519	78906	1	9	3,7,10,12	25061	516	13953	306	39836	2
6	2,3,10,13	18392	4499	2471	15911	41273	2	9	3,8,10,13	3029	18	1725	168	4940	1
6	2,4,8,15	44768	15282	42383	2701	105134	1	10	2,4,5,9	843	67592	2067	6503	77005	8
6	2,4,9,14	22436	853	4834	22793	50916	1	10	2,4,5,14	688	155085	5760	13006	174539	4
6	2,4,11,13	23569	922	1769	29068	55328	6	10	3,4,6,11	102	2399	8	164	2673	78
6	2,5,7,13	11252	894	3044	10945	26135	4	10	4,5,6,7	66389	3431	678	5263	75761	4
6	2,7,8,12	6744	757	4793	13076	25370	2	10	4,5,6,15	58359	1525	1234	5263	66381	4
6	2,7,8,15	20536	7641	4885	13076	46138	2	10	4,5,8,12	15380	550	941	151	17022	16
6	2,8,9,13	9620	98	3784	8101	21603	1	10	4,5,11,13	15563	364	492	1144	17563	5
6	2,8,10,14	162	110	49	59	380	33	10	4,7,9,11	22967	156	1352	1668	26143	2
6	2,8,12,13	1388	590	1892	757	4627	2	10	4,8,11,15	17657	104	1076	1196	20033	2
6	2,8,12,14	1732	337	1617	1514	5200	1	10	4,11,14,15	3123	192	26	208	3549	8
6	2,9,13,15	2088	676	1843	98	4705	1	11	1,3,5,12	265	7257	36613	35711	79846	4
6	2,12,14,15	4536	1977	5036	337	11886	1	11	2,3,5,9	373	67	1019	829	2288	132
7	1,2,4,14	4432	26	8779	2337	15574	28	11	2,4,5,14	22503	6553	82862	62621	174539	4
7	1,3,4,10	678	10	151	1297	2136	8	11	2,5,6,9	541	13490	8844	6698	29573	1
7	1,4,6,9	75367	143525	6344	27906	253142	1	11	2,5,6,15	21	323	194	197	735	34
7	1,4,6,13	8006	15830	55	4651	28542	6	11	2,5,8,10	3511	13381	7879	3834	28605	28
7	1,4,10,13	30	29	33	8	100	10	11	2,5,12,13	3067	23571	10347	12688	49673	2
7	1,4,10,14	64	43	91	10	208	8	11	2,5,13,15	4723	14241	3298	10347	32609	2
7	1,4,10,15	152	44	278	5	479	16	11	3,5,6,7	1676	83567	49814	30595	165652	1
8	1,2,3,7	17633	28610	171431	22934	240608	2	11	3,5,8,12	826	4874	265	4683	10648	4
8	1,2,3,13	20655	26750	157849	45868	251122	1	11	4,5,6,15	238	39299	28782	20189	88508	3
8	1,2,3,15	11419	10214	51009	22934	95576	2	11	5,6,9,10	119571	88052	53191	541	261355	1
8	1,2,5,10	66745	3274	41539	49561	161119	12	11	5,6,10,15	103633	75898	714	53191	233436	1
8	1,3,6,9	2193	14585	5988	1030	23796	5	11	5,6,12,14	3785	1124	1749	1406	8064	5
8	1,3,6,12	2616	17890	5176	2575	28257	2	11	5,8,12,15	11307	2407	9968	826	24508	4
8	1,3,11,14	4523	2132	5932	9799	22386	4	12	1,2,4,5	3861	7214	8827	2875	22777	92
8	1,5,6,13	24028	16925	4483	61335	106771	4	12	1,2,9,10	467	5591	7309	2418	15785	28
8	1,5,7,11	3284	2091	4838	1964	12177	14	12	2,3,4,14	6064	3551	3779	4228	17622	4
8	1,5,12,14	4225	2788	1263	10679	18955	16	12	2,4,7,11	4201	1046	6290	2977	14514	26
8	1,7,11,15	646	205	748	861	2460	34	12	2,4,8,9	4886	2792	3343	3678	14699	4
8	1,11,13,15	9778	10804	6970	11603	39155	1	12	2,4,8,15	16672	12154	16385	7356	52567	2
9	1,3,4,5	1926	43091	24609	2082	71708	4	12	2,5,7,10	19372	6561	30758	4706	61397	10
9	1,3,10,15	69	5116	3003	474	8662	4	12	2,8,10,11	28610	35593	28094	1370	93667	4
9	3,4,5,14	14479	7377	294	1926	24076	4	12	2,9,10,13	17111	14023	6497	13076	50707	1
9	3,4,7,15	1844	906	75	639	3464	4	12	2,9,10,14	3527	2265	1514	3269	10575	4
9	3,4,8,11	1495	771	78	48	2392	32	12	2,10,13,15	26536	11836	29826	14023	82221	1

[3] G. Szekeres and L. Peters, Computer solution to the 17-point Erdős–Szekeres problem, *ANZIAM J.* **48** (2006), 151–164, doi:10.1017/S1446181110000300X.

[4] F. Marić, Fast formal proof of the Erdős–Szekeres conjecture for convex polygons with at most 6 points, *J. Automat. Reason.* **62** (2019), 301–329, doi:10.1007/s10817-017-9423-7.

[5] A. Suk, On the Erdős–Szekeres convex polygon problem, *J. Amer. Math. Soc.* **30** (2017), 1047–1053, doi:10.1090/jams/869.

[6] M. J. H. Heule and M. Scheucher, Happy ending: an empty hexagon in every set of 30 points, in *Tools and Algorithms for the Construction and Analysis of Systems (TACAS 2024)*, Lecture Notes in Comput. Sci. **14570**, Springer, 2024, 61–80, doi:10.1007/978-3-031-57246-3_5.

[7] M. Scheucher, A SAT attack on Erdős–Szekeres numbers in $\mathbb{R}^d$ and the empty hexagon theorem, *Comput. Geom. Topol.* **2**(1) (2023), Article 2, 1–13, doi:10.57717/cgt.v2i1.12.

[8] G. Károlyi, Ramsey-remainder for convex sets and the Erdős–Szekeres theorem, *Discrete Appl. Math.* **109** (2001), 163–175, doi:10.1016/S0166-218X(00)00234-1.

[9] G. Károlyi and P. Valtr, Point configurations in d-space without large subsets in convex position, *Discrete Comput. Geom.* **30** (2003), 277–286, doi:10.1007/s00454-003-0009-4.

[10] C. Pohoata and D. Zakharov, Convex polytopes from fewer points, *Duke Math. J.* **174** (2025), 449–471, doi:10.1215/00127094-2024-0034.

[11] P. Valtr, Sets in $\mathbb{R}^d$ with no large empty convex subsets, *Discrete Math.* **108** (1992), 115–124, doi:10.1016/0012-365X(92)90665-3.

[12] L. Danzer, B. Grünbaum and V. Klee, Helly's theorem and its relatives, in *Convexity*, Proc. Sympos. Pure Math. **7**, Amer. Math. Soc., 1963, 101–180, doi:10.1090/pspum/007/0157289.

TABLE VI. The relations behind Theorem B, continued. Each row gives $\sum_j m_j x_{s_j} = M x_p$ with $\sigma = \{s_1, \dots, s_4\}$ in increasing order, positive integer weights m_j with sum M, and $|\det \sigma| = gM$ for the 4×4 determinant of the rows $(1, x_s)$.

p	σ	m_1	m_2	m_3	m_4	M	g	p	σ	m_1	m_2	m_3	m_4	M	g
12	3,4,6,15	64	468	2782	399	3713	8	14	1,7,9,11	107	492	410	262	1271	78
12	4,5,6,7	8542	2774	55354	9091	75761	4	14	1,8,9,10	493	7390	1956	2944	12783	4
12	4,5,6,13	9726	1279	54191	9091	74287	4	14	1,9,12,13	915	7412	1942	544	10813	1
12	4,6,8,11	2174	9752	1251	416	13593	13	14	1,10,12,15	861	714	654	4012	6241	28
12	4,6,9,11	1771	11336	1807	205	15119	9	14	2,3,7,15	1308	7224	8445	6181	23158	4
12	4,6,13,15	1236	7742	128	1279	10385	4	14	2,3,11,13	68	1839	446	25292	27645	4
12	5,6,9,10	657	196364	29492	34842	261355	1	14	3,4,7,8	639	1351	584	4262	6836	4
12	6,7,10,11	96074	13359	17426	8176	135035	1	14	3,5,7,12	9500	1213	12690	4103	27506	4
12	6,7,10,15	3492	73	626	511	4702	16	14	3,5,8,13	674	185	483	16388	17730	2
12	6,8,10,15	3370	146	688	365	4569	8	14	3,6,7,15	6366	2616	7263	5047	21292	2
12	6,9,10,11	29872	4453	5313	219	39857	3	14	3,6,9,13	3	1	2	29	35	366
13	1,2,4,14	5485	1118	12107	199326	218036	2	14	3,8,10,13	143	526	185	1616	2470	2
13	1,3,4,12	14660	62357	37121	29606	143744	2	14	3,8,13,15	18	171	1686	185	2060	2
13	1,4,8,9	2617	14795	35938	5246	58596	1	14	3,12,13,15	168	57	1638	61	1924	6
13	1,4,8,14	3	14	26	61	104	86	14	4,8,9,13	127	652	258	1580	2617	1
13	1,6,10,14	1111	433	2612	34331	38487	4	14	5,7,8,15	153	1137	789	335	2414	8
13	1,6,10,15	17930	9491	21574	68662	117657	2	14	7,8,9,10	986	4766	1250	1217	8219	2
13	1,7,9,11	7247	46781	25215	19895	99138	1	14	7,8,9,11	4690	4173	2981	1217	13061	2
13	1,7,12,15	2236	7328	3305	17285	30154	4	14	7,8,12,15	1113	1281	153	1465	4012	8
13	1,9,10,11	32261	37064	46781	28706	144812	1	14	8,9,11,13	1133	1051	127	9380	11691	1
13	2,3,7,11	5213	53574	110580	23621	192988	1	15	1,2,5,9	149	1025	2593	39087	42854	26
13	2,7,8,15	816	16327	12479	16516	46138	2	15	1,5,9,12	102	1039	13976	1025	16142	26
13	2,7,10,14	566	10230	707	102737	114240	1	15	1,9,11,12	269	85895	13507	615	100286	2
13	3,5,7,12	8302	1144	14043	4017	27506	4	15	2,3,4,11	949	59735	28841	14709	104234	4
13	3,7,8,10	28	95	291	100	514	52	15	2,5,9,14	927	2721	38973	1937	44558	2
13	3,7,12,15	86	111	49	44	290	104	15	2,8,9,11	151	608	56698	8669	66126	2
13	4,7,12,14	58	1771	309	17496	19634	2	15	3,4,5,7	2303	1039	866	1894	6102	16
13	5,6,7,14	71	149	1557	11252	13029	4	15	3,4,5,13	9661	5505	2266	7576	25008	4
13	5,7,8,9	335	2307	1104	68	3814	8	15	3,5,7,10	4024	1376	3031	2078	10509	8
13	6,7,8,15	204	1783	1367	1780	5134	8	15	3,5,8,10	6597	1129	3031	5225	15982	8
13	6,7,14,15	45	333	2734	71	3183	4	15	3,5,9,12	34	83	1188	97	1402	78
13	7,8,9,12	3839	4183	5296	1340	14658	2	15	3,6,9,11	551	534	57097	8691	66873	1
13	7,8,11,14	1051	215	192	4504	5962	1	15	3,10,11,12	38783	21151	8684	1618	70236	2
13	7,11,12,14	1813	116	215	14582	16726	1	15	5,6,8,9	4157	3404	567	61068	69196	2
14	1,2,7,15	7224	3180	17781	61453	89638	4	15	5,7,9,13	487	747	2788	394	4416	2
14	1,3,5,10	5297	144	3022	8242	16705	16	15	5,9,12,14	399	5040	309	442	6190	6
14	1,3,7,11	437	2665	7462	2802	13366	12	15	9,10,11,14	12492	3	2022	469	14986	2
14	1,4,5,11	1731	2378	1066	32	5207	52	15	9,11,13,14	945	153	6	29	1133	1
14	1,4,6,15	18987	15234	11388	94354	139963	2								
14	1,5,7,9	5041	10218	39908	332	55499	2								
14	1,5,13,15	288	225	12462	1207	14182	4								

[13] T. Bisztriczky and H. Harborth, On empty convex polytopes, *J. Geom.* **52** (1995), 25–29, doi:10.1007/BF01406823.

[14] T. Bisztriczky and V. Soltan, Some Erdős–Szekeres type results about points in space, *Monatsh. Math.* **118** (1994), 33–40, doi:10.1007/BF01305772.

[15] J. Schönheim, On coverings, *Pacific J. Math.* **14** (1964), 1405–1411, doi:10.2140/pjm.1964.14.1405.

[16] G. M. Ziegler, *Lectures on Polytopes*, Grad. Texts in Math. **152**, Springer, 1995, doi:10.1007/978-1-4613-8431-1.

[17] B. Grünbaum, *Convex Polytopes*, 2nd ed., Grad. Texts in Math. **221**, Springer, 2003, doi:10.1007/978-1-4613-0019-9.

[18] D. Bhattacharjee, U. Bhattacharya and S. Bhattacharya, The sphere packing problem in dimension 4, *Preprints* (2026), doi:10.20944/preprints202606.0856.v11.

[19] D. Bhattacharjee, U. Bhattacharya and S. Bhattacharya, The sphere packing problem in dimension 7, *Preprints* (2026), doi:10.20944/preprints202606.1534.v9.

[20] D. Bhattacharjee, Seed universality for reflexive polytopes in dimension four: generation of the Kreuzer–Skarke Calabi–Yau landscape via toric seed orbits, *Preprints* (2026), doi:10.20944/preprints202603.1056.v1.

[21] D. Bhattacharjee and U. Bhattacharya, Anaxia, the BLLM engine: verifier-gated conjecture closure, release v3.0.5 (2026), doi:10.5281/zenodo.23469389 (all versions doi:10.5281/zenodo.23286641), with the paper *BLLM: Beyond the LLM with Deep Human-Mind-AI*, `https://github.com/creelie/eBLLM`.

[22] D. Bhattacharjee and U. Bhattacharya, New lower bounds for Erdős–Szekeres numbers in three and four dimensions: coordinates, certificates and checks, repository `https://github.com/creelie/Erdos-s1`, release v1.0.0 (2026).